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# Rural electrification, migration and structural transformation: Evidence from Ethiopia☆

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ABSTRACT

To what extent does rural electrification induce structural change and alter migration patterns? This paper addresses this question using a simple multi-sector spatial model and evidence from a panel of rural Ethiopian villages during its recent expansion of electricity supply. We find that electrification raised irrigation rates, agricultural yields and non-agricultural business activity. Furthermore, electrified villages experienced higher rates of in-migration and lower rates of out-migration. Each of these predictions is qualitatively consistent with the predictions of our model. Our results suggest that rural electrification helped facilitate structural transformation of villages economies in Ethiopia and slowed out-migration from rural areas.

## 1. Introduction

Many economists and policymakers view rural electrification as a key driver of *structural transformation* in the developing world, meaning the movement of workers out of subsistence agriculture and into more productive activities. In this view, residents of rural areas without electricity must either eke out a living in subsistence agriculture or migrate to a city for an uncertain chance to obtain an urban wage job (as in Harris and Todaro, 1970). Once electrified, however, rural areas may gain productive opportunities for investments in agricultural enterprises or in non-farm businesses, both of which reduce the need for residents to migrate. Broad evidence in line with this view comes from Lipscomb et al. (2013), who find large effects of new hydroelectric dams on long-term regional economic outcomes in Brazil, and from Gaggl et al. (2019), who find a sizable impacts of electrification on non-agricultural business activity in U.S. counties in the early 20th century.

An alternative view is that rural electrification primarily improves the quality of village life, through better lighting and access to more consumer durables, rather than improving production opportunities. A growing body of research on rural electrification supports this view, beginning with Dinkelman (2011), who finds that electrification increases home appliance use in South African villages, but has only small effects on employment. More recently, Burlig and Preonas (2016) find that rural electrification in India increases energy use but without any impact on employment outcomes or income levels. Similarly, experimental evidence by Lee et al. (2020) reveals that rural Kenyans who receive electricity access use more electric light and own more televisions, but have no significant changes in earnings.

This paper examines the effects of rural electrification in Ethiopia, which made huge investments in electricity generation capacity in the last two decades. As of the year 2000, generation capacity for all of Ethiopia was a mere 500 MW, or around the capacity of a single medium-sized power plant in the United States. Effectively, no part of the country other than the capital of Addis Ababa had power. Twenty years later, Ethiopia has increased its generation capacity by a factor of ten, and become one of Africa's leaders in per capita electricity generation. No other country in the world had such a large growth rate of generation capacity over this period. Ethiopia's massive efforts to electrify rural areas, and the almost total concentration of rural employment in subsistence agriculture, make it especially well suited to study the relationship between rural electrification and structural transformation.

To guide our empirical analysis, we develop a simple multi-region model of how electrification of one region affects employment, productivity and consumption in that region, and how it effects migration

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patterns to and from other regions. The model builds on other structural spatial analyses of development programs that focus on worker mobility, such as Ahlfeldt et al. (2015) and Bryan and Morten (2019). Households in the model are born in one region, but receive a vector of region-specific productivity draws governing their productivity in all other regions. Households then choose their region and whether to work in the non-agricultural sector, the traditional-agricultural sector, or the modern-agricultural sector. The non-agricultural and modern-agriculture sectors both use electricity. Electricity is valuable because it increases consumption opportunities and because it raises productivity in both sectors. The model delivers simple closed-form expressions for sectoral and regional employment shares, among other outcomes.

The model's main comparative statics center on how electrifying one region affects that region and other regions of the economy in general equilibrium. There are four main predictions. First, electrifying a region will lead to structural change within agriculture, as capital and labor inputs move from traditional to modern agriculture. Second, electrifying a region raises consumption of electricity-using durable goods. Third, electrification raises a region's non-agricultural consumption and productivity. Fourth, electrifying one region reduces out-migration from that region and increases in-migration from all other regions, as households respond to the improved opportunities in the newly electrified region.

We test the predictions of the model using data from a nationally representative panel of Ethiopian villages between 2012 and 2014. To be sure, this is only a small window in which to observe Ethiopia's multi-decade efforts to electrify its rural areas. Though even during these few years, a substantial number of the villages in our data became electrified for the first time. We focus on the villages that were not electrified at all in 2012. We then document how villages that became electrified between 2012 and 2014 ("electrified villages") differed from those that still were not electrified at all in 2014 ("non-electrified villages.")

We begin by comparing the electrified and non-electrified villages in 2012, before any of the villages had access to electricity. Across two dozen observable characteristics, the two groups of villages are statistically indistinguishable on all but one. Both groups have similar populations, and are located equally far on average from Addis Ababa. Most villages reported net migration out, rather than in, and very few households in either village engage in any economic activity outside of agriculture. Using all observables jointly to predict which villages subsequently get electrified results in a negative adjusted $R^2$ and joint insignificance of all the predictors.

We then test the model's predictions by comparing how electrified and non-electrified villages changed between 2012 and 2014, the period in which the electrified villages received their grid connections. We find that electrified villages were more likely to begin irrigating in agriculture – which is made substantially easier with electric pumps. Electrified villages also reported higher agricultural yields. Both of these findings are consistent with the model's prediction of structural change within agriculture. When looking at ownership of electricity-using durables, we find that households in electrified villages were more likely to purchase televisions over this period. However, the magnitude of the change in television ownership is quite small, and other electricity-using durables have statistically insignificant differences in ownership rates.

Consistent with the model's predictions, we find evidence that non-agricultural productivity rose relatively more in electrified villages. In particular, households in electrified villages were about twice as likely to operate non-agricultural businesses by 2014, while non-electrified villages saw no such changes. These new businesses were most likely to be in retail and wholesale trade, with no significant differences in manufacturing and food processing, transportation or other non-agricultural businesses. Non-agricultural income rose significantly more in electrified villages (from a low initial level), and households were more likely to report subjective improvements in non-agricultural opportunities.

Turning to migration patterns, we find that in 2014, electrified villages were substantially less likely to report out-migration than non-electrified villages, and were somewhat more likely to receive migrants. These findings are consistent with our model's predictions. According to our model, the change in migration patterns could either be because opportunities for consumption of electricity-intensive durables increased in electrified villages, or because opportunities for production in the agricultural and non-agricultural sectors improved. Our empirical results suggest that the latter is the more promising channel, since the magnitudes of the estimated increases in productivity are much larger than the estimated increases in electricity-using durables. We conclude that rural electrification in Ethiopia likely played an important role in altering migration patters and driving structural change over this period.

Our work contributes to a growing literature on the role of rural electrification on development. Our findings sit more squarely among the papers that find substantial positive effects of electrification, such as Lipscomb et al. (2013) and Rud (2012), who find large long-run effects on regional development in Brazil and India, respectively. The empirical effects we find are larger in magnitude, though less precisely estimated, than those of Lee et al. (2020) and Burlig and Preonas (2016), who find negligible short-run impacts of rural electrification programs in Kenya and India. Our paper's focus on Ethiopia is shared by Moneke (2020), who finds positive effects of rural electrification, especially when it is combined with investment in roads. From a macro perspective, our work supports the role of infrastructure as a determinant of measured total-factor productivity (Prescott, 1998).[1]

Our paper also contributes to the large and still growing literature on the determinants of internal migration in developing countries, started by Harris and Todaro (1970). Our focus on electrification relates most closely in this literature to the papers by Dinkelman (2011), who finds evidence that electrification altered migration patterns in South Africa, and Dinkelman and Schulhofer-Wohl (2015), who show how the welfare effects of spatial investments, such as rural electrification programs, depend on migration effects. More broadly, our work contributes to the debate about the determinants of urban informality in developing-world cities (see e.g. Fields, 1975; Williamson, 1988; Lucas, 1997). Many policymakers worry about how the steady steam of rural-urban migrants strains urban infrastructure and urban labor markets. The results of this paper suggest that rural infrastructure provision may be a way to slow or even reverse rates of migration to cities. For recent reviews of the literature on rural-urban migration in developing countries see Brueckner and Lall (2015) and Lagakos (2020).

## 2. Ethiopia's electrification

Ethiopia began the new millennium as one of the countries with the lowest rates of rural electrification in the world. Fig. 1 plots Ethiopia's generation capacity, meaning its stock of electric power plants, in 1990, 2000 and 2014. We compute generation capacity using the World Electric Power Plants Database (WEPP), which collects generation capacity (in MW) on every electric power facility in the world above a threshold of 25 kW. We complement the WEPP database with additional information from various public sources since 2014.

In 2000, Ethiopia had just 441 MW of generation capacity, which was half the sub-Saharan African average of 846 MW. Given that Ethiopia is one of the largest countries in Africa, generation capacity per capita was negligible. Average power consumption per capita was

[1] Other studies in this literature have focused on the effects of electricity access on firm performance (Fisher-Vanden et al., 2015; Allcott, Collard-Wexler, and O'Connell, 2016; Abeberese, 2017; Abeberese et al., 2019; Fried and Lagakos, 2020) and human capital accumulation (Vidart, 2020), constraints on payment for electricity (Jack and Smith, 2015; Dzansi et al., 2018), and competition in electricity markets (Ryan, Forthcoming), among other topics.

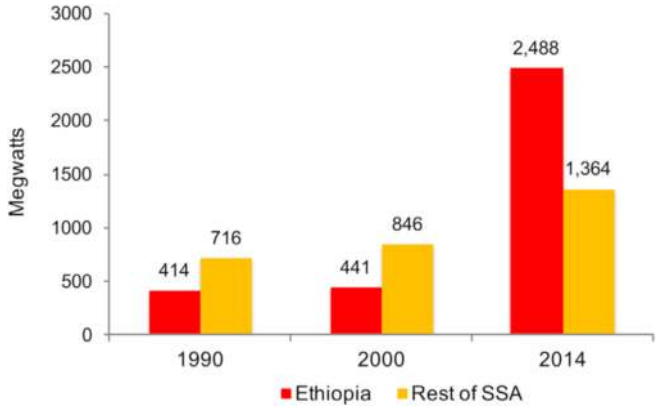


**Fig. 1.** Electricity Capacity: Ethiopia vs the Rest of Sub-Saharan Africa (SSA).

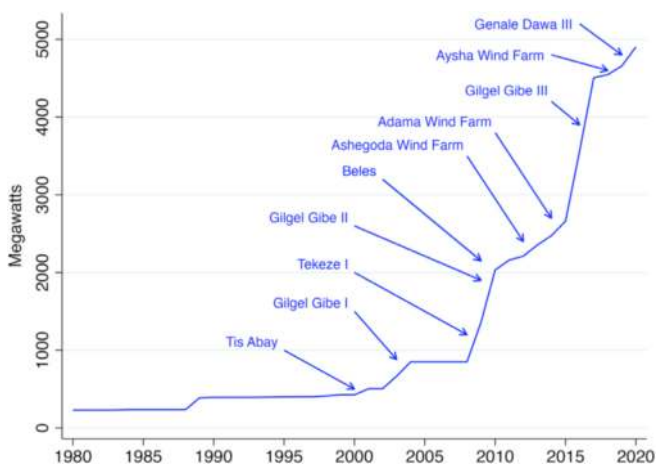


**Fig. 2.** Electricity capacity in Ethiopia.

21 kWhs a year, approximately equal to what the average U.S. resident consumes in one day (IEA, 2016). According to the World Bank, only 13 percent of rural Ethiopians had access to electricity.

By 2014, Ethiopia's generation capacity had jumped dramatically to 2488 MW, an increase by a factor of 5.6. Compared to the rest of Sub-Saharan Africa, which averaged 1364 MW, Ethiopia had almost double the electricity capital stock. In per capita terms, Ethiopia tripled its per capita consumption to 64 kWhs per year. The rate of rural electrification more than doubled, increasing from 13 percent to 27 percent.

Fig. 2 plots the times series of electricity generation capacity in Ethiopia, to provide a more granular look at Ethiopia's dramatic electrification over this period. Before 2000, the generation capacity barely changed, staying below 500 MW. After 2000, the country completed in rapid succession a series of hydroelectric plants, starting from Tis Abay, Gilgel Gibe I, Tekeze I, Gilgel Gibe II and Beles, all before 2010. In 2012 and 2013, it completed two large wind farms, Ashegoda and Adama. More recently, it completed the largest dam to date: Gilgel Gibe III, which helped bring its total generation capacity to almost 5000 MW in 2020. It seems safe to say that no country has ever increased its generation capacity by a factor of ten in just two decades, as Ethiopia has.

## 3. Model of electrification, migration and structural transformation

Given the massive scope of Ethiopia's electricity expansion, the impacts of electrification of one rural area are unlikely to be contained in that region. In this section, we present a simple multi-region, multi-sector model to understand how rural electrification affects migration patterns between regions and structural transformation within regions. The model makes several predictions at the regional level, which we then test using our panel of rural Ethiopian villages.

### 3.1. Households and geography

The economy is composed of $N$ regions, each of which is inhabited by a continuum of households. Each household $i$ is endowed with one unit of labor which they supply inelastically to the labor market in exactly one region. If household $i$ born in region $j$ supplies their labor in region $k$, they must pay an (iceberg) migration cost which reduces their utility by factor $\frac{1}{d_{kj}}$. Parameter $d_{kj}$ is increasing in the distance between regions $k$ and $j$. Furthermore, $d_{jj} = 1$ for all $j$, implying that households born in region $j$ that also choose to work in region $j$ do not pay a migration cost.

Following Roy (1951) and Eaton and Kortum (2002), each household $i$ is endowed with a vector of region-specific productivities, $\{\nu_j(i)\}_{j=1}^N$. Thus, household $i$'s efficiency units of labor in region $j$ are $\nu_j(i)$. A household's productivity in region $j$ is drawn from the region-specific Frechet distribution:

$$F_j(\nu_j) = \exp(-T_j \nu_j^{-\zeta}), \qquad T_j > 0 \quad \text{and} \quad \zeta > 1. \tag{1}$$

Household $i$ born in region $j$ draws their region-specific productivities at birth. Based on these draws, they must decide whether to supply their labor in region $j$ or to migrate and supply their labor in region $k \neq j$.[2]

Households split their consumption between an agriculture good, $c^a$, and a non-agriculture good, $c^n$, and electricity, $c^e$. Preferences are a Cobb-Douglas aggregate of these three goods:

$$U(c^a, c^e, c^n) = (c^a)^\gamma (c^e)^\eta (c^n)^{1-\gamma-\eta}, \tag{2}$$

where parameter $\gamma$ governs the relative taste for agriculture consumption and parameter $\eta$ governs the relative taste for electricity consumption. The Cobb-Douglas assumption is mainly for tractability; our model's comparative statics apply more broadly to any utility function where the three goods are normal.

There is no savings mechanism; households consume all of their income. The budget constraint for household $i$ in region $j$ is:

$$p_j^a c_j^a(i) + p_j^e c_j^e(i) + c_j^n(i) = y_j(i), \tag{3}$$

where $y_j(i)$ denotes the labor income of household $i$ working in region $j$. Variables $p_j^a$ and $p_j^e$ are the relative prices of the agriculture good and electricity in region $j$, respectively. The non-agriculture good is the numeraire and can be used to purchase both electricity and the agriculture good. We assume that the non-agriculture good can be traded across regions at zero cost, and therefore has the same price in every region. The household optimization yields the following standard demands for the agriculture, electricity, and non-agriculture goods, respectively:

$$c_j^a(i) = \frac{\gamma y_j(i)}{p_j^a}, \quad c_j^e(i) = \frac{\eta y_j(i)}{p_j^e} \quad \text{and} \quad c_j^n(i) = (1-\gamma-\eta) y_j(i). \tag{4}$$

[2] Our assumption of region-specific, rather than general, productivities is consistent with the study of Bazzi et al. (2016), who find evidence that skills of rural households in Indonesia are region-specific, based on the differences in crop types and climate characteristics. We abstract from education and fertility decisions, though these are relevant for migration decisions; see e.g. Liao et al. (2020).

### 3.2. Production

Perfectly competitive firms in each region $j$ produce an agriculture good, $Y_j^a$, and a non-agriculture good, $Y_j^n$, from labor (measured in efficiency units) and electricity. Firms purchase the electricity intermediate at exogenous price $p_j^e$. If the region is connected to the national electric grid, then firms purchase this electricity directly from the grid. If the region is not connected to the grid, firms can still purchase electricity from diesel generators and other off-grid sources. However, this off-grid electricity is considerably more costly than the grid electricity. The government has the option to "electrify" a region by connecting the region to the grid, and thus reducing the price of electricity in that region.[3]

#### 3.2.1. Non-agricultural production

The production technology for the non-agriculture good in region $j$ is Cobb-Douglas in labor, $L_j^n$, and electricity, $E_j^n$:

$$Y_j^n = A^n (E_j^n)^\theta (L_j^n)^{1-\theta}, \tag{5}$$

where $A^n$ denotes TFP in non-agriculture production (constant across regions) and $\theta$ is the factor share of electricity. The representative firm's first order conditions yield the firm's demand for electricity:

$$E_j^n = \left(\frac{\theta A^n}{p_j^e}\right)^{\frac{1}{1-\theta}} L_j^n, \tag{6}$$

and the equilibrium wage (per efficiency unit) of labor:

$$w_j^n = (1-\theta) A^n (E_j^n)^\theta (L_j^n)^{-\theta}. \tag{7}$$

To derive the production function and the wage as functions of the electricity price, substitute firm electricity demand (equation (6)) into the production function for non-agriculture (equation (5)) and also into the expression for the equilibrium wage (equation (7)) to get:

$$Y_j^n = (A^n)^{\frac{1}{1-\theta}} \left(\frac{\theta}{p_j^e}\right)^{\frac{\theta}{1-\theta}} L_j^n \quad \text{and} \quad w_j^n = (1-\theta)(A^n)^{\frac{1}{1-\theta}} \left(\frac{\theta}{p_j^e}\right)^{\frac{\theta}{1-\theta}}. \tag{8}$$

All else constant, equation (8) implies that electrifying a region (i.e. a reduction in the electricity price, $p_j^e$) increases the region's non-agriculture output and raises the equilibrium wage in the non-agriculture sector.

#### 3.2.2. Agriculture production

Perfectly competitive firms can produce the agriculture good using a traditional or a modern technology. Production with the traditional technology requires only labor, while production with the modern technology requires both labor and electricity. For example, traditional agriculture might rely on rainfall or small, gravity-driven aqueducts to water crops, while modern agriculture might instead use an irrigation system with an electric pump. The production functions in region $j$ for traditional agriculture, $Y_j^t$, and modern agriculture, $Y_j^m$, are:

$$Y_j^t = A^t L_j^t \qquad \text{and} \qquad Y_j^m = A^m (E_j^m)^\alpha (L_j^m)^{1-\alpha}, \tag{9}$$

where $A^t$ and $A^m$ denote total factor productivity (constant across all regions) in traditional and modern agriculture, respectively. Parameter $\alpha$ is the electricity share in the production of modern agriculture.

Following the same procedure as for the non-agriculture good, we derive the production function and the equilibrium wage for modern agriculture as a function of the electricity price:

$$Y_j^m = (A^m)^{\frac{1}{1-\alpha}} \left(\frac{\alpha p_j^a}{p_j^e}\right)^{\frac{\alpha}{1-\alpha}} L_j^m \quad \text{and}$$

$$w_j^m = (1-\alpha)(p_j^a)^{\frac{1}{1-\alpha}} (A^m)^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{p_j^e}\right)^{\frac{\alpha}{1-\alpha}}. \tag{10}$$

The firm's first order condition for labor yields the equilibrium wage in the traditional agriculture sector:

$$w_j^t = p_j^a A^t. \tag{11}$$

Agriculture goods produced with the traditional and the modern technologies are perfect substitutes. However, since the equilibrium wages in both sectors do not depend on labor (equations (10) and (11)), all households will choose to work for the producers that use the technology with the higher marginal product of labor, and thus pay a higher wage. Therefore, producers of the agriculture good in region $j$ will use either the traditional or the modern technology, but not both.

### 3.3. Within-region equilibrium

Households in each region demand positive quantities of both the agriculture and the non-agriculture goods (equation (4)). The model's equilibrium forces allocate households between agriculture and non-agriculture production to meet these demands. This allocation requires that households are indifferent between working in the agriculture and the non-agriculture sectors, and thus necessitates equal wages across the two sectors:

$$w_j^n = \max[w_j^t,\ w_j^m], \tag{12}$$

where the max operator reflects the fact that, in equilibrium, only the production technology with the highest marginal product of labor is used to produce the agricultural good.

Since wages (given by equations (7), (10) and (11)) are independent of labor in each sector, the price of the agriculture good adjusts in each region to ensure that equation (12) holds. This yields the following expression for the equilibrium price of agriculture in region $j$:

$$p_j^a = \begin{cases} (1-\theta)(A^n)^{\frac{1}{1-\theta}} \left(\frac{1}{A^t}\right) \left(\frac{\theta}{p_j^e}\right)^{\frac{\theta}{1-\theta}} & : p_j^e \geq \widetilde{p} \\ \left(\frac{1-\theta}{1-\alpha}\right)^{1-\alpha} (A^n)^{\frac{1-\alpha}{1-\theta}} \left(\frac{1}{A^m}\right) \left(\frac{\theta}{p_j^e}\right)^{\frac{\theta(1-\alpha)}{1-\theta}} \left(\frac{\alpha}{p_j^e}\right)^{-\alpha} & : p_j^e < \widetilde{p}, \end{cases} \tag{13}$$

where $\widetilde{p}$ is the value of the electricity price that equates the modern and the traditional wages in agriculture, $w_j^t = w_j^m$. Specifically, $\widetilde{p}$ is given by:

$$\widetilde{p} = \theta^\theta \alpha^{1-\theta} (1-\theta)^{1-\theta} (1-\alpha)^{\frac{(1-\theta)(1-\alpha)}{\alpha}} A^n \left(\frac{A^m}{A^t}\right)^{\frac{1-\theta}{\alpha}}. \tag{14}$$

Note that the cutoff value of the electricity price, $\widetilde{p}$, is the same across all regions (since TFP does not vary by region). If the electricity price is high, $p_j^e \geq \widetilde{p}$, then the traditional technology has the highest marginal product of labor, implying that all agricultural production will use the traditional technology. The reverse is true if the electricity price is low. Therefore, region $j$'s output of the agriculture good, $Y_j^a$, is given by:

$$Y_j^a = \begin{cases} Y_j^t & : p_j^e \geq \widetilde{p} \\ Y_j^m & : p_j^e < \widetilde{p}. \end{cases} \tag{15}$$

**Prediction 1** *Electrifying a region using traditional agriculture will cause the region to switch to modern agriculture as long as the decrease in the electricity price is sufficiently large.*

The intuition for this prediction is simply that if electrifying region $j$ reduces $p_j^e$ below the cutoff price, $\widetilde{p}$, then equation (15) implies that

[3] We abstract from trade since rural Ethiopia is largely a closed economy. See e.g Stokey (2001), Uy et al. (2013), Tombe (2015), Bustos et al. (2016) and Teignier (2018) for open-economy models of structural change, in settings where food trade plays an important role.

region $j$ will switch from traditional to modern agriculture. Additionally, note that advances in modern agriculture technology, $A^m$, increase the electricity-price cutoff, $\widetilde{p}$, implying that the region will switch to the modern agricultural production technology at a higher value of the electricity price. In equilibrium, wages are equated across all sectors in a region, implying that the wage in region $j$, $w_j$, equals the wage in the non-agriculture sector:

$$w_j = w_j^n = (1-\theta)(A^n)^{\frac{1}{1-\theta}}\left(\frac{\theta}{p_j^e}\right)^{\frac{\theta}{1-\theta}}. \tag{16}$$

Equation (16) thus reveals that electrifying region $j$ increases its equilibrium wage which raises labor income:

$$y_j(i) = w_j v_j(i), \tag{17}$$

for all households in region $j$. This increase in labor income combined with the decrease in the electricity price raises household demand for electricity.

**Prediction 2** *Electrifying a region increases household consumption of electricity.*

To see this prediction, differentiate the demand for electricity by household $i$ in region $j$ (equation (4)) with respect to the electricity price in region $j$:

$$\frac{\partial c_j^e(i)}{\partial p_j^e} = \underbrace{-\eta y_j(i)\left(\frac{1}{p_j^e}\right)^2}_{\text{Substitution effect}} + \underbrace{\left(\frac{\eta}{p_j^e}\right)\frac{\partial y_j(i)}{\partial p_j^e}}_{\text{Income effect}} < 0 \tag{18}$$

Decreases in the electricity price increase electricity demand through both income and substitution effects. The substitution effect (first term in equation (18)) occurs because the decrease in the relative price of electricity causes households to substitute the consumption of electricity for both agriculture and non-agriculture consumption. The income effect (second term in equation (18)) arises because the decrease in the electricity price increases wages in region $j$, raising household income and, thus, increasing demand for electricity, as well as for the agriculture and non-agriculture goods.

**Prediction 3** *Electrifying a region increases non-agriculture production per household.*

The intuition for this result is that electrifying a region increases labor income (equation (17)), which raises household demand for the non-agriculture good (equation (4)), which, in turn, increases its supply per household.

### 3.4. Inter-region equilibrium

We next solve for the equilibrium allocation of households across the $N$ regions. A household born in region $j$, will migrate to region $k$ if their utility in region $k$, net of their migration costs, is greater than their utility in region $j$. To solve for the inter-region equilibrium, we must first solve for household utility in each region. Substitute the demand functions for the agriculture, electricity, and non-agriculture goods (equation (4)) into the utility function (equation (2)) to derive the utility of a household, $i$, born region $j$ as a function of labor income and relative prices:

$$U_j(i) = y_j(i)\left(\frac{\gamma}{p_j^a}\right)^{\gamma}\left(\frac{\eta}{p_j^e}\right)^{\eta}(1-\eta-\gamma)^{1-\eta-\gamma}. \tag{19}$$

Electrifying a region reduces the price of electricity, causing households to consume more of the electricity good, raising utility. Electrification also increases labor income, $y_j(i)$, which raises utility through the income effect. However, working in the other direction, electrification increases the relative price of the agriculture good, which, all else constant, decreases the household's consumption of the agriculture good, lowering utility. To determine which of these effects dominates, substitute the equilibrium price of agriculture (equation (13)), and household labor income (equations (16) and (17)) into equation (19) to obtain household utility as a function of the electricity price:

$$U_j(i) = H(p_j^e)v_j(i), \tag{20}$$

where:

$$H(p_j^e) = \begin{cases} \Theta(1-\theta)^{1-\gamma}(A^n)^{\frac{1-\gamma}{1-\theta}}(A^t)^{\gamma}\left(\frac{\theta}{p_j^e}\right)^{\frac{\theta(1-\gamma)}{1-\theta}}\left(\frac{\eta}{p_j^e}\right)^{\eta} & : p_j^e \geq \widetilde{p} \\ \Theta(1-\theta)^{1-\gamma(1-\alpha)}(1-\alpha)^{\gamma(1-\alpha)}(A^n)^{\frac{1-\gamma(1-\alpha)}{1-\theta}} & \\ (A^m)^{\gamma}\left(\frac{\theta}{p_j^e}\right)^{\frac{\theta-\gamma\theta(1-\alpha)}{1-\theta}}\left(\frac{\alpha}{p_j^e}\right)^{\gamma\alpha}\left(\frac{\eta}{p_j^e}\right)^{\eta} & : p_j^e < \widetilde{p}. \end{cases}$$

and $\Theta \equiv \gamma^{\gamma}(1-\eta-\gamma)^{1-\eta-\gamma}$.

$H'(p_j^e) < 0$, implying that electrifying region $j$ increases the utility of households in the region. Hence, the utility gains from higher labor income and electricity consumption dominate the utility losses from the lower agriculture consumption (income held constant).

We next solve for the ex-ante probability (i.e. before the region-specific productivities realize) that a household born in region $j$ chooses to supply their labor in region $k$. The utility of household $i$ born in region $j$ and working in region $k$ is:

$$U_{kj}(i) = H(p_k^e)\left(\frac{v_k(i)}{d_{kj}}\right).$$

The household will migrate to the region where they receive the highest utility net of migration costs. Taking this migration into account, the utility of a household $i$, born in region $j$ is:

$$U_j(i) = \max\{U_{kj}(i);\ k = 1, \ldots, N\}.$$

Since each household's labor productivity in region $j$ is the realization of a random variable, $v_j$, their expected utility in region $k$ is also the realization of the random variable, $U_{kj} = H(p_k^e)\left(\frac{v_k}{d_{kj}}\right)$. Moreover, the highest expected utility is the realization of the random variable, $U_j = \max\{U_{kj}: k = 1, \ldots, N\}$. The likelihood that a household $i$ born in region $j$ supplies their labor to region $k$ is the probability, $\pi_{kj}$, that region $k$'s utility turns out to be the highest.

Let $G_{kj}(U)$ be the CDF for the expected utility for households born in region $j$ and working in region $k$:

$$\begin{aligned} G_kj(U) &= \text{Prob}[U_kj \leq U] \\ &= \exp\left(-T_k\left(\frac{H(p_k^e)}{d_kj}\right)^{\zeta}U^{-\zeta}\right). \end{aligned} \tag{21}$$

We solve for the distribution of the highest utility available to a given household born in region $j$, $G_j(U)$. This utility will be greater than $U$ unless the utilities in every region are less than $U$:

$$\begin{aligned} G_j(U) &= \text{Prob}\left[U_j \leq U\right] \\ &= \prod_{k=1}^{N} G_kj(U) \\ &= \exp\left(-U^{-\zeta}\sum_{k=1}^{N}\psi_kj\right) \quad \text{where} \quad \psi_kj = T_k\left(\frac{H(p_k^e)}{d_kj}\right)^{\zeta}. \end{aligned} \tag{22}$$

Finally, we solve for the probability that a household from region $j$ chooses to work in region $k$, $\pi_{kj}$. A household from region $j$ will migrate to region $k$ if the utility in region $k$ is greater than the utility in all the other regions:

$$\pi_k j = \text{Prob}\left[U_k j \geq \max_{s \neq k} U_s j\right]$$
$$= \int_0^\infty \prod_{s \neq k} G_s j(U) dG_k j(U).$$
$$= \frac{\psi_k j}{\sum_{s=1}^N \psi_s j} \tag{23}$$

Equations (22) and (23) imply that the probability that a household from region $j$ migrates to region $k$ depends, in part, on the electricity price in region $k$. This observation leads to the following proposition.

**Prediction 4** *Electrifying a region raises in-migration and lowers out-migration.*

To see that electrification increases in-migration, observe that $\frac{\partial \pi_{kj}}{\partial p_k^e} < 0$. Decreases in the electricity price in region $k$ increase the probability that households migrate from region $j$ to region $k$. Similarly, to see that electrification reduces out-migration, observe that $\frac{\partial \pi_{jj}}{\partial p_j^e} < 0$. Decreases in the electricity price in region $j$ increase the probability that households in region $j$ choose to work in region $j$.

### 3.5. Numerical illustration

We use a simple numerical example to illustrate the implications of Prediction 4. We set the Fréchet parameter, $T_j$ equal to 1 for all $j$ and the Fréchet parameter $\zeta$ equal to 2. Drawing on data for South Africa from the World Bank's Global Consumption Database, we set the household's expenditure share on electricity, $\gamma$, equal to 0.03 and the household's expenditure share on the agriculture good, $\eta$, (equivalent to the food and beverage category in the data) equal to 0.59. The factor share of electricity in both non-agriculture, $\theta$, and modern agriculture, $\alpha$, production equals 0.07, as in Allcott, Collard-Wexler, and O'Connell (2016). TFP in traditional agriculture, $A^t$, equals 1, and TFP in modern agriculture, $A^m$, equals 1.1, implying that agriculture firms receive a 10 percent productivity boost from using the modern technology. TFP in non-agriculture production, $A^n$, equals 1. Finally, we set the migration cost from migrating from any region $j$ to any region $k \neq j$ equal to 5, $d_{kj} = 5$.

We construct a two-period experiment to highlight the effects of electrification on migration. We divide the economy into three regions, $A$, $B$, and $C$. A unit mass of households is born into each region in each period. In period 1, region A is electrified with electricity price $p^e = 0.99\widetilde{p}$. Regions $B$ and $C$ are not electrified and have electricity price $p^e = 5\widetilde{p}$. We make this choice given the evidence of Foster and Steinbuks (2008) that self-produced electricity from a generator costs around five times as much as electricity from the grid on average.

The left bar in each region-block of Fig. 3 shows the number of households who choose to work in that region in period 1. The shades correspond to the household's birth region, with light, medium and dark shaded bars representing households born in regions $A$, $B$, and $C$, respectively. Since region $A$ is electrified in period 1 and regions $B$ and $C$ are not, almost all households born in region $A$ stay in region $A$ (the height of the light segment of the leftmost-bar in Fig. 3 is approximately one) while one quarter of households born in regions $B$ and $C$ choose to migrate to region $A$ (the heights of the light and dark segments of the leftmost bar in Fig. 3 are each approximately 0.25). There is a small amount of migration between regions $B$ and $C$ in period 1, even though neither region is electrified. The migrants from region $B$ to region $C$ correspond to the households that have high labor-productivity draws for region $C$ and low productivity draws for regions $A$ and $B$. Similarly, the migrants from region $C$ to region $B$ have high labor-productivity draws for region $B$ and low draws for regions $A$ and $C$.

To illustrate how Prediction 4 works, we electrify region $B$ in period 2. Regions $A$ and $C$ are the same in period 2 as they were in period 1 (electrified and not-electrified, respectively). The right bar in each region-block of Fig. 3 shows the number of households who choose to work in each region in period 2. Again, the shaded segments correspond to the household's birth region. Prediction 4 implies that electrifying Region $B$ should reduce out-migration. We can see this result in Fig. 3 by comparing the height of the medium-shaded segment of the bar in the region-$B$ block when region $B$ is electrified (period 2) with the height when region $B$ is not electrified (period 1). The medium-shaded segment is larger in period 2, implying that when region $B$ is electrified, more people born in region $B$ choose to work in region $B$.

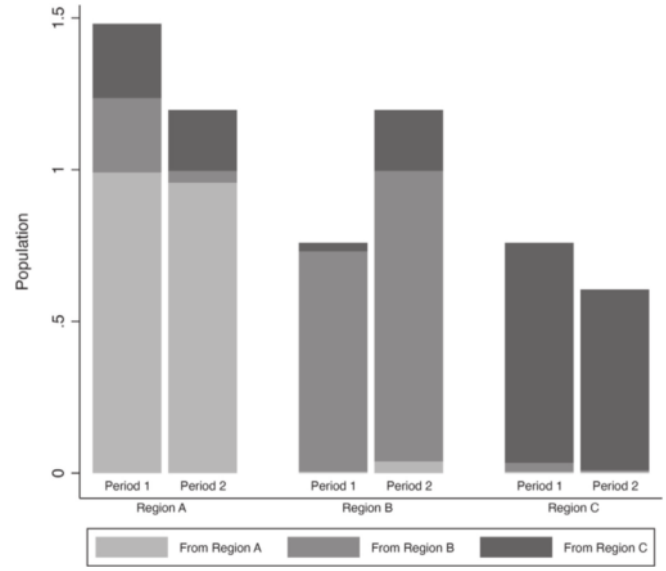


**Fig. 3.** The effects of electrification on migration patterns: An Illustration.

Prediction 4 also implies that electrifying region $B$ will increase in-migration to that region. We can see this result in Fig. 3 by comparing the heights of the light and dark segments of the bars in the region-$B$ block when region $B$ is electrified (period 2) with their heights when region $B$ is not electrified (period 1). The dark and light segments are larger in period 2 than in period 1, implying that when region $B$ is electrified, more people migrate to region $B$ from regions $A$ and $C$. There are more in-migrants from region $C$ than from region $A$ because region $C$ is not electrified.

Migration implies that electrification of a subset of regions will have spillover effects on the other regions. The migration-induced changes in population are the most direct form of these spillovers. For example, in Fig. 3, electrifying region $B$ reduces the population in regions $A$ and $C$. These spillovers can also affect production levels in non-electrified regions. In the numerical example above, electrifying region $B$ reduces non-agriculture production in region $C$ because region $C$'s population decreases. These cross-region spillovers imply that one cannot infer the effects of a nationwide electrification policy by scaling up the effects from electrifying a sub-set of regions. Such an exercise would overestimate the benefits from nationwide electrification because some of the electrified regions' gains occur because of the non-electrified regions' losses.

### 3.6. Summary of Model's predictions

Predictions 1–4 summarize four results of the model that we can test directly using a differences-in-differences analysis of electrified and non-electrified villages in Ethiopia. Prediction 1 implies that electrifying a village causes a shift from traditional to modern agriculture. In our empirical work, we will look at productivity and technology use, broadly defined, in farming. Prediction 2 posits that electrifying a village increases household consumption of electricity. Increases in household electricity consumption could imply that households are using more light and/or that households purchased electricity-using

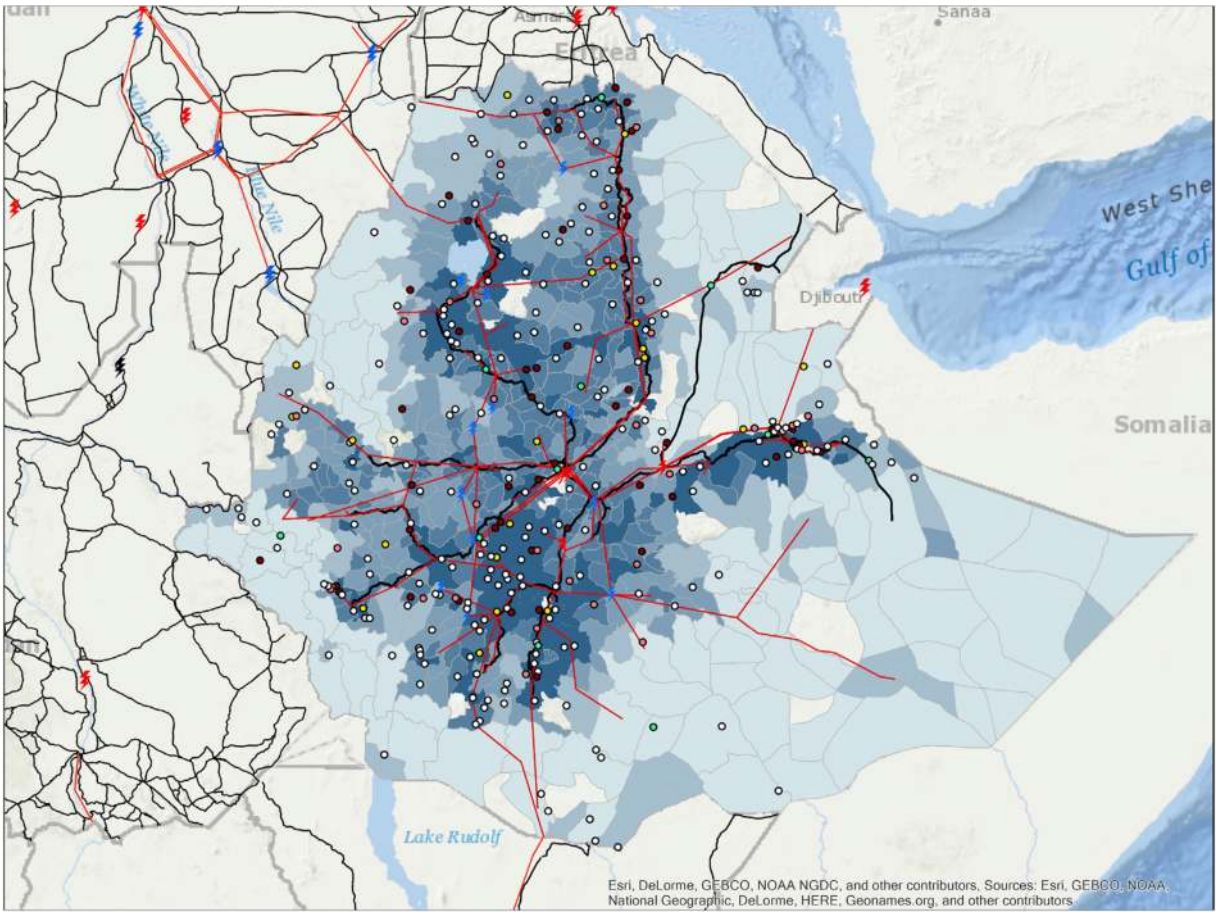


**Fig. 4.** Ethiopian population density and ERSS sample villages.

durables. We will test the latter channel, and look at household use of electricity-using durables, such as electric stoves, sewing machines and televisions. Prediction 3 shows that electrifying a village increases the production of the non-agriculture good per household. We can test this directly by analyzing the effects of electrification on the fraction of households that have a non-agriculture business, and the profitability of non-agricultural businesses. Prediction 4 posits that electrifying a village will increase in-migration and decrease out-migration. We can also directly test this prediction by analyzing the effects of electrification on whether a village is more likely to be a net-sender or a net-receiver of migrants.[4]

## 4. Data

We now describe how we test the predictions of the model using a representative panel of rural Ethiopian villages from 2012 to 2014.

### 4.1. Ethiopian Rural Socioeconomic Survey

Our main data source is the Ethiopian Rural Socioeconomic Survey (ERSS) conducted in 2011/2012 and again in 2013/2014 on a set of enumeration areas representative of parts of Ethiopia. The former survey was representative of all rural areas of Ethiopia, while the latter was nationally representative. Usefully for our purposes, most of the enumeration areas surveyed in 2012 were also surveyed in 2014. Fig. 4 shows the location of sample enumeration areas in the ERSS, along with population density (darker regions being more dense), major roads (thicker lines), and the high-voltage electricity grid (thinner, straighter, lines); it is clear that the survey coverage is representative of Ethiopia's populated areas. We restrict attention to rural enumeration areas (henceforth "villages") that were surveyed twice, so as to take advantage of the panel structure. We further restrict attention to villages that are at least 25 km from Addis Ababa and have no more than 10,000 persons, to make sure our sample is truly of small rural communities and not suburban areas or cities.

Since our focus is on electrification, we restrict attention to villages in which none of the respondents report having access to grid electricity in 2012. This allows us to compare villages that became connected to the grid in 2014 to those that remain unconnected in 2014. We end up with a nationally representative panel of 215 rural Ethiopian villages that are not electrified as of 2012.

The period 2012 to 2014 was a period of large increases in electricity production and distribution in Ethiopia. At a national level, we calculate that total electricity production capacity in Ethiopia grew from 2159 MW in 2011 to 1475 by the end of 2014, a 15 percent increase in capacity. Moreover, there was a substantial push to electrify urban areas besides Addis Ababa. Along the way, the Ethiopian Electric Utility, the government agency in charge of power distribution, made a commitment to electrify rural villages throughout the country.

Several government sources suggest that equity was the main criteria used to decide which rural villages to electrify. Ethiopia's recent

[4] Our model leaves out the important of issue of housing prices, which may adjust in response to migration. See Brueckner and Zenou (1999) and Brueckner and Lall (2015) for formal treatments of housing in Harris-Todaro style models. We also ignore the issue of who pays for the infrastructure investments, leaving a full cost-benefit analysis of this large infrastructure investment future work. The fact that we find positive effects on agricultural and non-agricultural productivity suggests that the investment pays for itself in part.

**Table 1**
Average village characteristics in 2012.

| | Village Type | | |
|---|---|---|---|
| | Not Electrified | Electrified by 2014 | Difference (*p*-val) |
| Population | 4574 | 4925 | 350 (0.38) |
| Dist. to Major Road (km) | 53 | 46 | −7 (0.46) |
| Dist. to Addis (km) | 337 | 326 | −11 (0.63) |
| Dist. to Addis via road (km) | 513 | 462 | −51 (0.14) |
| Dist. to Djibouti via road | 962 | 895 | −67 (0.19) |
| Dist. to Electrified E.A. (km) | 44 | 38 | −6 (0.49) |
| Dist. to Line From Dam to Addis | 114 | 93 | −21 (0.22) |
| Health center (%) | 18.1 | 27.9 | 9.8 (0.15) |
| Dist. to School (km) | 0.8 | 2.0 | 1.2 (0.13) |
| Irrigation Scheme (%) | 47.1 | 46.5 | −0.5 (0.53) |
| Farmers Irrigating | 126.2 | 160.5 | 34.3 (0.25) |
| Received Migrants (%) | 23.2 | 20.9 | −2.3 (0.75) |
| Sent Migrants (%) | 43.0 | 55.8 | 12.8 (0.13) |

Source: Authors' Calculations Using Ethiopian Rural Socioeconomic Survey (2012, 2014).

Universal Access Plan states that the primary goal of rural electrification is to bring electricity access to all rural areas. In practice, the central government provides quotas to each region on the number of villages they can choose to electrify, and the regions decide on the list of individual villages.[5]

### 4.2. Non-electrified and electrified villages

We focus on comparisons of two groups: "non-electrified villages", which we define as villages with no access to grid electricity in either year, and "electrified villages," or those that did not have grid electricity in 2012 but got access by 2014. To classify villages into one of these two groups, we use household-level questions on access to grid electricity. Formally, non-electrified villages are those households in which exactly zero percent of households report having grid electricity in both years. Electrified villages have at least one household that reports having grid electricity in 2014. The rationale is that moving from zero to a positive fraction of households with grid electricity likely reflects an expansion of the electricity grid, rather than increased demand for electricity by village residents. Of the 215 villages in our sample, we classify 172 villages as non-electrified and 43 villages as electrified.

Our main empirical approach is to compare the change in average characteristics between 2012 and 2014 in the electrified villages and the non-electrified villages:

$$(\overline{y}_{e,14} - \overline{y}_{e,12}) - (\overline{y}_{c,14} - \overline{y}_{c,12}), \tag{24}$$

for a variety of average outcomes, $\overline{y}_{it}$. For expositional purposes we also report the changes in the electrified villages, $\overline{y}_{e,14} - \overline{y}_{e,12}$, and non-electrified villages, $(\overline{y}_{c,14} - \overline{y}_{c,12})$ separately. In Appendix A, we complement this approach with fixed-effect regressions of electrification on various outcomes, which provide a nearly identical set of results.

Table 1 presents average characteristics of the two types of villages. The population (first row) of non-electrified villages averages 4,574, compared to 4925 for electrified villages. The third column reports the difference, which is 350, and the *p*-value from a two-sided *t*-test of the null hypothesis that the difference is zero, which is 0.38. Thus, populations are not statistically different for the two village types. Notice that our villages are far larger than the set of Indian villages studied by Burlig and Preonas (2016), which have between 150 and 450 people, and the rural Kenyan communities studied by Lee et al. (2020), which have 85 households per community on average.

The next four rows of Table 1 present various measures of geographic isolation, including distance to a major road, distance to Addis Ababa as the bird flies, distance to Addis Ababa via roads, and distance to the port of Djibouti via roads. In short, both sets of villages are approximately equally isolated, being 53 and 46 km from the nearest major road, for example, and 512 and 462 km from Addis via road. None of the differences in distances are statistically significant.

The next two rows report two measures that may be informative about the cost of electrification. The first reports the distance in kilometers to the nearest electrified enumeration area that has at least half of its households reporting access to grid electricity in 2012. Non-electrified villages are 44 km away, compared to 38 km for villages that later get electrified, a statistically insignificant difference. The second measure reports the distance in kilometers to a straight line running from the nearest hydroelectric dam to Addis Ababa. Proximity to this line may imply a higher chance of being electrified, since the cost of electrifying villages closer to this line is lower (Moneke, 2020). In our set of villages, electrified villages are indeed closer to this line: the average distance to electrified villages is 93 km, compared to 114 km in the non-electrified villages. Yet the *p*-value of the difference is 0.22, making it statistically insignificant from zero at conventional significance levels. Thus, for this set of villages, there is not strong evidence that proximity to existing power lines is a main factor in electrification decisions.

The bottom six rows of Table 1 compare the two village groups by their access to public goods, current agricultural practices and migration patterns. The public goods access is similar in the two regions, with both non-electrified and electrified villages having a modest 18.1 and 27.9 percent with access to health clinics, respectively, and the distance to the nearest school being 0.8 and 2.0 km. Irrigation schemes in agriculture are also similar in the two groups of villages, with similar fractions and numbers of farmers irrigating their fields. The migration characteristics are also similar, which may be informative about other unobservables at the village level. Among non-electrified villages, 23.2 percent had more migrants moving in than out, compared to 20.9 percent among electrified villages. Around 43 percent of non-electrified villages and 55.8 percent of electrified villages report having more out-migrants than in-migrants. Neither difference is statistically significant, and all in all, both groups are net senders of migrants, which is consistent with Ethiopia's rapidly rising rates of urbanization over this period.

To summarize the village level comparisons, we find no significant differences in 2012. Still, households within the villages could differ in important ways. Table 2 reports average characteristics of household respondents in the ERSS. The first four rows report the average percent of households with access to tap water, a flush toilet, having a mud

[5] In a private meeting with the CEO of the Ethiopian Electric Utility, we asked which criteria the regional governments use to select rural villages for electrification. His answer was equity, rather than cost or village growth potential, though the Utility and Ethiopian Electric Power Company often advise regional governments to consider cost as well, to the extent possible.

**Table 2**
Average household characteristics by village type in 2012.

| | Village Type | | |
|---|---|---|---|
| | Not Electrified | Electrified by 2014 | Difference (*p*-val) |
| Tap Water (%) | 9.6 | 8.9 | −0.7 (0.86) |
| Flush Toilet (%) | 2.0 | 0.2 | −1.8 (0.36) |
| Mud Floor (%) | 96.9 | 94.8 | −2.1 (0.14) |
| Use Generator (%) | 23.5 | 22.6 | 0.9 (0.88) |
| Non-agric. business (%) | 4.9 | 5.4 | 0.6 (0.65) |
| Monthly Non-ag Sales (100s of Birr) | 3.2 | 4.4 | 1.2 (0.19) |
| Mobile Phone (%) | 19.9 | 25.9 | 6.1∗ (0.09) |
| Electric Stove (%) | 1.0 | 1.9 | 0.9 (0.41) |
| Sewing Machine (%) | 1.5 | 2.4 | 0.9 (0.46) |
| Television (%) | 1.6 | 2.8 | 1.1 (0.32) |

Source: Authors' Calculations Using Ethiopian Rural Socioeconomic Survey (2012, 2014)

floor and using a generator for lighting. In both village types, less than ten percent of households have access to tap water or toilets, and nearly all have a mud floor. Around one quarter of households use a generator for their lighting. None of the differences are statistically significant at the ten percent level or lower.

The next two rows of Table 2 report the average fraction of households reporting that they operate a non-agricultural business of any kind, and the average monthly sales of non-agricultural goods. Just 4.9 percent of non-electrified villagers and 5.4 percent of villagers whose villages are later electrified operate non-agricultural businesses. Average sales were 320 and 440 Birr in the two village types. Neither difference is statistically significant. The final four rows of Table 2 report ownership of electricity-using durable goods. Two features are worth noting here. First, mobile phone ownership rates are fairly high for both villages, at 19.9 percent and 25.9 percent, with a statistically significant difference between them (at the ten percent level). Next, for all the other durable goods, we see no significant differences.

In sum, across twenty three metrics, all but one show similar characteristics with statistically insignificant differences in 2012 between villages that got electrified by 2014 and those that did not. The lack of differences between village types in 2012 is also consistent with the direct account of the Ethiopian Electric Utility that reports equity as being the primary factor in deciding which villages got electrified over this period, rather than estimated potential for future growth. Still, it is possible that the two groups of villages are on different trends, even if they appear similar in levels in 2012. Unfortunately, the panel of villages does not extend back before 2012, making it impossible to check directly for parallel trends.

As an additional approach to assessing determinants of electrification by 2014, we take a forecasting approach and attempt to predict electrification using the variables in Tables 1 and 2 for 2012. In the first regression, we include just the village-level characteristics. In the second, we combine the village and household characteristics. Table A.1 reports the results. None of the variables besides the constant are significant, and the adjusted $R^2$ is −0.11 in the first regression and −0.009 in the second. The *F*-stat of joint significance of all independent variables is 0.90 with a *p*-value of 0.59. We conclude that future electrification is not easily predicted in 2012, using the available observables. This is consistent with equity being the primary consideration in deciding which rural villages to electrify, rather than observable village characteristics.[6]

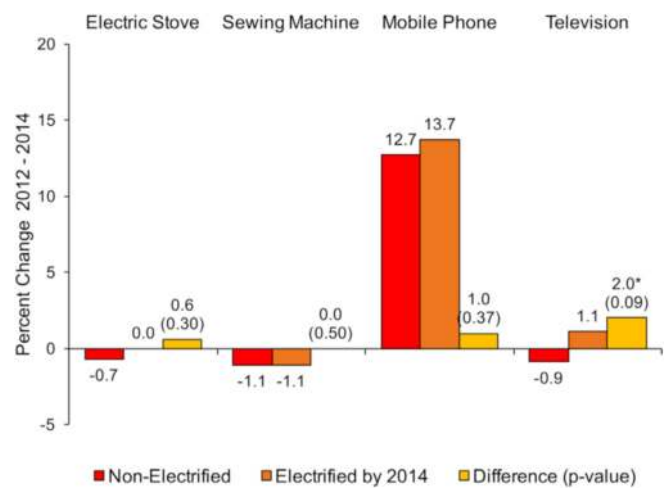


**Fig. 5.** Differences in household durable good ownership.

## 5. Empirical findings

In this section, we compare the characteristics of villages that became electrified by 2014 to those that remained non-electrified. In particular, we assess the model's predictions for the effects of electrification on consumption of electricity-intensive goods, the extent of non-agricultural activities within the village, agricultural productivity, (electricity-using) irrigation rates, and migration patterns. We also ask whether other public goods were extended contemporaneously along with the electricity grid.[7]

### 5.1. Consumption of electricity intensive goods

Our main findings concern the difference in economic outcomes from 2012 in 2014, in electrified villages compared to differences over the same period in non-electrified villages. In other words, we use a differences-in-differences approach to estimate the effects of rural electrification. The simplest prediction of the model is that electrification leads to increases in the consumption of electricity, or electricity-using durables. Using the ERSS, data we can compute changes in the ownership rates of four main electricity-using durable goods: electric stoves, sewing machines, mobile phones and televisions.

Fig. 5 plots changes in durable ownership from 2012 to 2014. The red bars plot the average changes for the non-electrified villages, the

[6] The fact that a village is electrified does not, of course, mean that electricity is reliable, in the sense that there are no power outages. Nor does it mean that all households in the electrified villages are actually hooked up to the grid. In the electrified villages, on average, 18.8 percent of surveyed households actually reported having a grid connection in 2014.

[7] In Appendix Tables A:2 through A:6 we also present the results of regressions that provide a very siimilar set of results to the ones presented in this section.

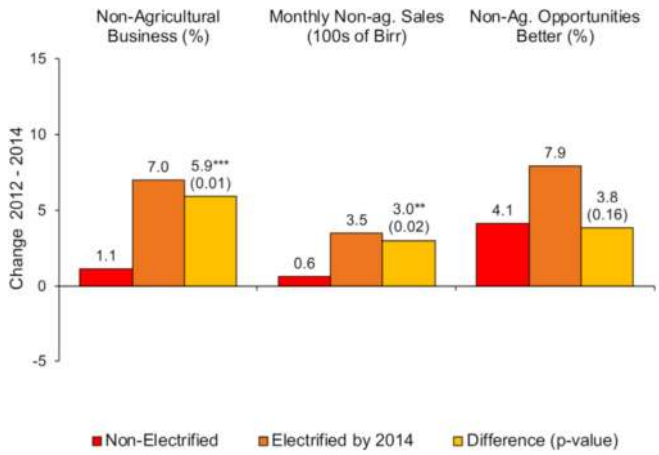


**Fig. 6.** Changes in non-agricultural activity.

orange bars plot the average changes for the electrified villages and the yellow bars are the simple difference (with *p*-value in parenthesis). As the Figure shows, changes in ownership of stoves and sewing machines were negligible across both village groups. Changes in mobile phones were fairly dramatic, at a 12.7 percent and 13.7 percent increase in the two village groups. The difference, however, was statistically insignificant.

The one place where we find a significant difference in ownership rate increases is in televisions. Non-electrified villages have a decrease of 0.9 percent in ownership rates, while electrified villages have an increase of 1.1 percent. The difference is 2 percent, which is statistically significant at the ten percent level (*p*-value 0.09). Still, this is a fairly modest difference in economic terms. We conclude that the pure consumption effects of electrification are positive, as predicted by the model, but small.

### 5.2. Non-agricultural businesses

Our model predicts that electrification increases non-agricultural production. The best simple proxies for non-agricultural production in the ERSS is the data on the operation of non-agricultural businesses and on the monthly sales and revenues from those businesses, conditional on operation. Another measure of changes in non-agriculture production comes from the village-level questionnaires, which include a question on whether respondents feel that non-agricultural opportunities have improved in recent years.

Fig. 6 reports the results for non-agricultural production. The fraction of households with an agricultural business increased by just 1.1 percent in non-electrified villages compared to 7 percent in electrified villages, for a difference of 5.9 percent, which is significant at the one-percent level. This amounts to roughly doubling the number of non-agricultural businesses in the electrified villages, a substantial increase. Monthly, non-agricultural sales also increase significantly more (300 Birr) in the electrified villages. Finally, electrified villages were 3.8 percentage points more likely to report that non-agricultural opportunities improved, but the *p*-value for this difference is 0.16.

Table 3 reports the results by industry, to understand which types of businesses were most responsive to electrification. Our data allow a breakdown into five major sectors: manufacturing and food processing, retail and wholesale trade, restaurants and hotels, transportation and other non-agricultural businesses. We find that most new businesses in electrified villages are in retail and wholesale trade. Ownership of retail and wholesale trade business declines by 1.1 percent in non-electrified villages but increases by 6.5 percentage points in electrified villages. The difference in the two changes is statistically significant at the one percent level. The increase in ownership of retail and wholesale trade businesses in electrified villages represents more than doubling of the pre-electrification ownership rates in 2012 (less than 5 percent in most villages).

Note that, as discussed in Section 3 above, the model predicts that the higher non-agricultural production in electrified villages could in part come from lower production in non-electrified villages. These spillovers are consistent with the observed increase in retail business activity in the electrified villages and decline in the non-electrified villages. However, the magnitudes suggest that the overall difference of 7.6 percentage points is unlikely to be driven by a simple shift of retail activity across villages, since the decline in the non-electrified villages (−1.1 percent) is much smaller than the gain (+6.5 percent) in the electrified villages.

Ownership rates in other industries do not change much with electrification. Manufacturing and food processing is one of the goals of the Ethiopian government, particularly in rural areas, where food processing is a natural sector to expand given the predominance of agriculture. Still, there was essentially no change in the operation of manufacturing or food processing businesses in either non-electrified or electrified villages. Restaurants are also a natural business to open after electrification, though there are also no effects on that sector. One possibility is that entrepreneurs open restaurants which also offer retail sales, and as a result, are classified mostly as retail establishments. Its not obvious why transportation businesses would expand, but we find no changes there or for other miscellaneous non-agricultural businesses.

### 5.3. Agricultural production

The model predicts that electrification will increase agricultural productivity. Fig. 7 reports the percent differences from 2012 to 2014 in agricultural variables in non-electrified villages (red bars), electrified villages (orange bars) and their difference (yellow bars). As above, the number in parentheses below the difference is the *p*-value from the null hypothesis that the true difference is zero.

The first 3 bars cover the changes in the percent of villages that have an irrigation scheme, where farmers collectively attempt to irrigate village farmland. Non-electrified villages saw 15.3 percent more irrigation schemes, while electrified villages saw 25.6 percent more. Electrified villages were therefore 10.3 percent more likely to have an irrigation

**Table 3**
Differences in non-agricultural business by industry.

| | Percentage Point Changes 2012–2014 | | |
|---|---|---|---|
| | Non-Electrified | Electrified in 2014 | Difference |
| Manufacturing | −0.5 | −0.3 | 0.2 (0.94) |
| Retail and Wholesale Trade | −1.1 | 6.5 | 7.6*** (0.01) |
| Restaurants and Hotels | 0.1 | −0.2 | −0.3 (0.69) |
| Transportation | 4.7 | 2.5 | −2.2 (0.80) |
| Other non-agricultural Business | 0.6 | 0.4 | −0.1 (0.52) |

Source: Authors' calculations Using Ethiopian Rural Socioeconomic Survey (2012, 2014). *p*-values are in parentheses in the Difference column. ***, **, and * mean statistically significant at the 1-, 5- and 10-percent levels.

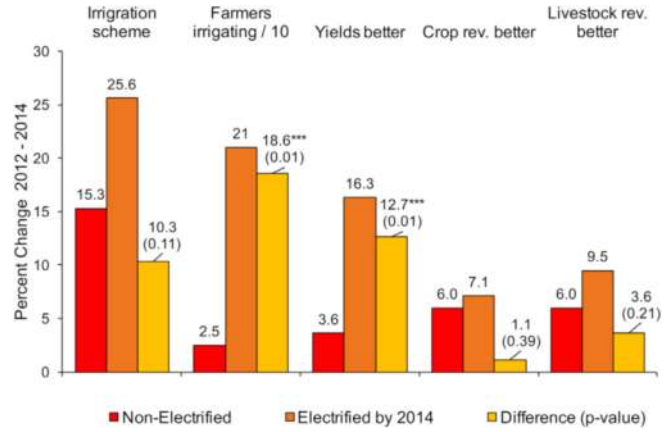


**Fig. 7.** Changes in agricultural production.

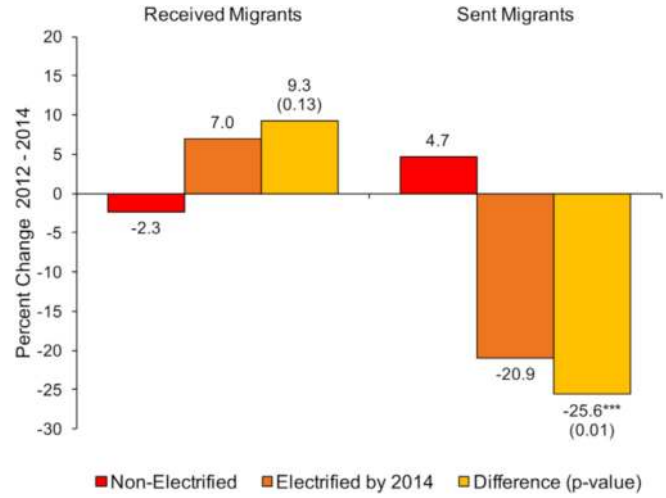


**Fig. 8.** Changes in migration patterns.

scheme, a difference which is marginally statistically significant (with a *p*-value of 0.11). The next 3 bars report the number of farmers participating in irrigation schemes (divided by ten, for expositional purposes). Electrified villages saw 210 more farmers participating, compared to just 25 for non-electrified villages, for a highly significant difference of 185 more farmers irrigating.

How did the irrigation affect farm productivity and revenues? The third set of bars show the fraction of villages reporting that yields were much better in 2014 than 2012. A substantial 16.3 percent of electrified villages report having much higher yields, compared to 3.6 percent in non-electrified villages, representing a highly significant difference of 12.7 percent. This higher productivity in farming could be the direct result of the increased irrigation, though the evidence here cannot rule out other mechanisms. For example, Caunedo and Keller (forthcoming) find that adoption of the latest vintages of tractors is a central determinant of cross-country differences in agricultural productivity. It is possible that there are complementarities between tractor adoption and irrigation. Unfortunately, we cannot measure tractor usage in our data. A related issue is whether our short time span is long enough to observe the full effects of technology adoption in response to electricity. Tractors are known to have long adoption delays for example (Manuelli and Seshadri, 2014), and one would need a longer time period than ours to pick up the full effects on adoption.

### 5.4. Migration patterns

The model predicts that migration patterns change in response to electrification. Fig. 8 shows how in-migration and out-migration patterns changed in practice in the panel of villages. The first set of bars reports changes in the fraction of villages reporting that they received more migrants than they sent. Non-electrified villages are 2.3 percent less likely to receive migrants, while electrified villages are 7.0 percent more likely, for a difference of 9.3 percent. Though only marginally significant, this points in the direction of electrified villages becoming more likely to be receivers of migrants.

The second set of bars reports changes in the fraction of villages sending more migrants than they receive. Non-electrified villages are 4.7 percent more likely to be net senders of migrants, while electrified villages are a dramatic 20.9 percent less likely to net senders of migrants. The difference is 25.6 percent, and statistically significant at well below the one percent level. In summary, village migration patterns are clearly associated with electrification. Electrified villages are somewhat more likely to receive migrants, and substantially less likely to send them. This pattern is just as the model predicts.

**Table 4**
Differences in other public goods.

| | Percentage Point Changes 2012–2014 | | |
|---|---|---|---|
| | Non-Electrified | Electrified in 2014 | Difference |
| Distance to school (km) | 0.3 | −1.5 | −1.8 (0.10) |
| Health center (%) | 3.0 | −7.0 | −10.0 (0.92) |
| Tap water (%) | 3.5 | 14.4 | 10.9** (0.02) |
| Flush Toilet (%) | −0.5 | 0.8 | 1.3 (0.28) |

Source: Authors' Calculations Using Ethiopian Rural Socioeconomic Survey (2012, 2014)

### 5.5. Other public goods

Finally, we look at whether other public goods were introduced simultaneously in villages that became electrified over this period. If so, one may not want to conclude that any of the above effects were the result of electrification itself, but rather the result of multiple new public goods that were introduced at the same time. We focus on four public goods that we can measure in the ERSS in both years: distance to schools, whether there is a health clinic in the village, whether the village has tap water, and whether there is a sewer system, proxied for by flush toilet ownership.[8]

Table 4 reports the differences in these public goods rates across the two village types between 2012 and 2014. Schools and health clinics had similar and statistically insignificant changes over this period, as did flush toilets. Interestingly, villages that became electrified had significantly larger changes in tap water access over the period: non-electrified villages had 3.5 percent increases in access, compared to 14.4 percent increases on average in electrified villages.

One interpretation of this finding is that, as outlined above, village electrification efforts coincided with efforts to increase access to tap water. In this case, one should interpret all the results in the previous section as the effects of receiving both electricity and tap water. Another interpretation, which we find more plausible, is that tap water access itself is the result of electrification. In Ethiopia, local governments are in charge of tap water provision, while electrification is undertaken by the (national) Ethiopian Electric Power company and the Ethiopian

[8] Moneke (2020) makes the compelling case that electricity and all-weather roads are highly complementary, and often launched in tandem. Unfortunately our data do not have a comparable measure of tar road access in the two years of our study.

Electric Utility. So any increases in tap water would have been independent of any electrification, at the village level. More directly, electrification would allow for electric pumps to pump water from wells, which would make it easier to increase tap water access. While it is hard to prove this order of cause and effect, the political independence of local governments and national power utilities suggests tap water and electricity are independent decisions, and the reverse direction of causality (tap water to electricity) seems fairly implausible.

## 6. Conclusion

To what extent does rural electrification alter migration patterns and induce structural transformation? We answer this question both theoretically and empirically. We build a two-sector model with many regions and individual migration decisions that builds on spatial models of structural change. We test the model's predictions using a panel of rural Ethiopian villages that were not yet electrified in 2012. We then compare villages that got electrified by 2014 to those that remained non-electrified. As the model predicts, we find that electrifying a village leads to a modest increase in consumption of electricity-using durables, increases in agricultural productivity – likely working through irrigation – and increases in non-agricultural activity and business income. Electrified villages experienced substantial decreases in out-migration, relative to non-electrified villages, and modest increases in in-migration.

The pioneering paper by Harris and Todaro (1970) was the first to treat migration patterns in the developing world as the result of optimizing behavior by rational individuals. The current paper advances the research agenda that Professors Harris and Todaro started by linking internal migration decisions to rural electrification and structural transformation. It is well known that the costs of rural electrification are high in developing countries, particularly in per capita terms. This paper provides some evidence that the benefits may also be substantial. Electricity connections in Ethiopian villages helped promote structural transformation of agricultural and non-agricultural production. Rural households responded by voting with their feet, and were more likely to remain in villages with "bright lights" than in those left in the dark.

## Declaration of competing interest

None.

## Appendix

**Table A.1**
Predicting Electrification by 2014.

| | | |
|---|---|---|
| Population | 7.53e-06 | 0.000013 |
| | (0.000013) | (0.000013) |
| Dist. to Major Road (km) | 1.88e-07 | 1.44e-07 |
| | (8.63e-07) | (9.27e-07) |
| Dist. to Djibouti via Road (km) | −0.000078 | −0.000083 |
| | (0.00011) | (0.00012) |
| Dist. to Addis via Road (km) | −0.00015 | −0.00019 |
| | (0.00017) | (0.00019) |
| Dist. to Electrified EA | −8.74e-08 | −2.60e-07 |
| | (9.82e-07) | (1.02e-06) |
| Sent Migrants, pct | 0.077 | 0.067 |
| | (0.067) | (0.069) |
| Received Migrants, pct | 0.049 | −0.022 |
| | (0.080) | (0.086) |
| Flush Toilet, pct | | −0.19 |
| | | (0.26) |
| Mud Floor, pct | | −0.46 |
| | | (0.36) |
| Tap Water, pct | | 0.0026 |
| | | (0.13) |
| Use Generator, pct | | 0.090 |
| | | (0.096) |
| Non-agricultural business | | 0.059 |
| | | (0.35) |
| Sold agricultural product | | 0.34 |
| | | (0.23) |
| Tar Road, pct | | −0.00056 |
| | | (0.078) |
| Health Center, pct | | 0.10 |
| | | (0.073) |
| Electric Stove, pct | | −0.0075 |
| | | (0.98) |
| Sewing Machine, pct | | −0.48 |
| | | (1.00) |
| Mobile Phone, pct | | 0.23 |
| | | (0.15) |
| Television, pct | | 0.71 |
| | | (1.01) |
| Constant | 0.26** | 0.61 |
| | (0.13) | (0.38) |
| Observations | 213 | 213 |
| Adjusted $R^2$ | −0.011 | −0.009 |

Source: Authors' Calculations Using Ethiopian Rural Socioeconomic Survey (2012, 2014). Sample is all rural villages that are not electrified in 2012 and have non-missing values for all variables. Dependent variable is electrified by 2014. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

**Table A.2**
Electrification and Durable Goods Ownership.

| VARIABLES | (1) Elec Stove | (2) Sew. Machine | (3) Mobile | (4) TV |
|---|---|---|---|---|
| Electrified | 0.0089 | 0.0084 | 0.061 | 0.011 |
| | (0.0083) | (0.0085) | (0.038) | (0.0094) |
| 2013 | −0.0066 | −0.011** | 0.13*** | −0.0090 |
| | (0.0053) | (0.0054) | (0.024) | (0.0059) |
| Electrified∗2013 | 0.0061 | −0.000053 | 0.0100 | 0.020 |
| | (0.012) | (0.012) | (0.053) | (0.013) |
| Constant | 0.010*** | 0.015*** | 0.20*** | 0.016*** |
| | (0.0037) | (0.0038) | (0.017) | (0.0042) |
| Observations | 430 | 430 | 430 | 430 |
| Adjusted $R^2$ | 0.006 | 0.009 | 0.086 | 0.025 |

Standard errors in parentheses.
*** p < 0.01, ** p < 0.05, * p < 0.1.

**Table A.3**
Electrification and Agricultural Productivity.

| VARIABLES | (1) Irrigation Scheme | (2) Number Irrigating |
|---|---|---|
| Electrified | −0.0055 | 34.3 |
| | (0.084) | (61.6) |
| 2014 | 0.15*** | 26.2 |
| | (0.053) | (39.2) |
| Electrified∗2014 | 0.11 | 193** |
| | (0.12) | (87.9) |
| Constant | 0.47*** | 126*** |
| | (0.038) | (27.7) |
| Observations | 427 | 423 |
| Adjusted $R^2$ | 0.026 | 0.032 |

Standard errors in parentheses.
*** p < 0.01, ** p < 0.05, * p < 0.1.

**Table A.4**
Electrification and Non-Agricultural Productivity

| VARIABLES | (1) Bus | (2) Sales | (3) Mfg Bus | (4) Trade Bus | (5) Hot Rest Bus | (6) Oth Bus | (7) Trans Bus |
|---|---|---|---|---|---|---|---|
| Electrified | 0.0064 | 116 | −0.0054 | 0.0094 | 0.0039 | −0.0028 | 0.034* |
| | (0.019) | (144) | (0.024) | (0.027) | (0.0042) | (0.017) | (0.019) |
| 2013 | 0.012 | 62.2 | −0.0052 | −0.011 | 0.0015 | 0.0062 | 0.049*** |
| | (0.012) | (91.2) | (0.015) | (0.017) | (0.0026) | (0.011) | (0.012) |
| Electrified∗2013 | 0.058** | 293 | 0.0023 | 0.076** | −0.0032 | −0.0018 | −0.024 |
| | (0.027) | (204) | (0.034) | (0.038) | (0.0059) | (0.024) | (0.027) |
| Constant | 0.048*** | 320*** | 0.065*** | 0.078*** | 0.0039** | 0.049*** | 0.013 |
| | (0.0086) | (64.6) | (0.011) | (0.012) | (0.0019) | (0.0076) | (0.0086) |
| Observations | 429 | 429 | 429 | 429 | 429 | 429 | 429 |
| Adjusted $R^2$ | 0.030 | 0.018 | −0.007 | 0.018 | −0.005 | −0.006 | 0.039 |

Standard errors in parentheses.
*** p < 0.01, ** p < 0.05, * p < 0.1.

**Table A.5**
Electrification and Migration.

| VARIABLES | (1) Send Migrants | (2) Receive Migrants |
|---|---|---|
| Electrified | 0.13 | −0.023 |
| | (0.085) | (0.072) |
| 2013 | 0.047 | −0.023 |
| | (0.054) | (0.045) |
| Electrified∗2013 | −0.26** | 0.093 |
| | (0.12) | (0.10) |
| Constant | 0.43*** | 0.23*** |
| | (0.038) | (0.032) |
| Observations | 430 | 430 |
| Adjusted $R^2$ | 0.004 | −0.005 |

Standard errors in parentheses.
*** p < 0.01, ** p < 0.05, * p < 0.1.

**Table A.6**
Electrification and Public Goods Fixed.

| VARIABLES | (1) Dist School | (2) Hospital | (3) Tap Water | (4) Flush Toilet |
|---|---|---|---|---|
| Electrified | 1.18 | 0.098 | −0.0070 | −0.019 |
| | (1.00) | (0.069) | (0.045) | (0.017) |
| 2013 | 0.27 | 0.036 | 0.038 | −0.0048 |
| | (0.64) | (0.044) | (0.028) | (0.011) |
| Electrified∗2013 | −1.82 | −0.11 | 0.11* | 0.013 |
| | (1.42) | (0.098) | (0.063) | (0.024) |
| Constant | 0.77* | 0.18*** | 0.096*** | 0.020*** |
| | (0.45) | (0.031) | (0.020) | (0.0075) |
| Observations | 426 | 427 | 429 | 429 |
| Adjusted $R^2$ | −0.003 | −0.002 | 0.017 | −0.004 |

Standard errors in parentheses.
*** p < 0.01, ** p < 0.05, * p < 0.1.

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